

JOBFAIR AI

*Detecting Inflated Skill Requirements in Indian Tech Job Postings
using SQL, Python & Generative AI*

Data Science Project Competition — July 2026

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1. Problem Statement

Job postings in the Indian tech industry frequently list inflated requirements — asking for years of experience and long skill lists for what are, in practice, entry-level roles. A posting labelled ‘Junior Data Analyst’ will often demand 3–5 years of experience and half a dozen tools, discouraging genuinely qualified candidates from applying.

“Why does an entry-level posting ask for 5 years of experience?” — the question this project set out to answer with real data, rather than anecdote.

This mismatch creates false rejection anxiety among candidates, especially freshers and early-career professionals, who often self-reject from roles they are capable of performing. At the same time, recruiters face inefficiency from filtering irrelevant or over-qualified applications caused by unclear postings. Both sides lose from a market where job requirements do not reliably reflect the actual role.

This project systematically measures how widespread this gap is across real Indian tech job postings, using SQL for pattern discovery, Python for scoring, and a GenAI-powered chatbot to explain findings in plain language.

2. Overall Solution Approach

The project follows a four-stage pipeline, taking raw job posting data through to a live, explainable chatbot:

2.1 Data Collection & Cleaning

- Dataset: ~19,600 real Data Science / Data Analytics job postings scraped from Naukri.com (fields: job title, company, experience required, salary package, location, skills).
- Experience Required (e.g. ‘4–8 Yrs’) was parsed into numeric exp_min / exp_max columns using regex.
- The Skills column had no separators between individual skills (e.g. ‘Product managementCareer development’). A regex rule — inserting a boundary wherever a lowercase letter is directly followed by an uppercase letter — was used to split it into a clean list of skills per posting.
- Rows with critically missing data (no Skills or no Experience Required) were dropped, leaving ~19,229 usable postings.

2.2 Relational Database Design (MySQL)

Instead of a single flat file, the cleaned data was loaded into a normalized MySQL database with two related tables:

- postings — one row per job posting (title, company, exp_min, exp_max, location, skill_count)
- skills — one row per individual skill, linked back to its posting via a posting_id foreign key

This one-to-many structure enables real SQL joins and aggregations — for example, comparing skill demand across experience bands or identifying the most frequently requested skills — rather than simple spreadsheet filtering.

2.3 SQL Analysis

A series of analytical SQL queries were run directly against the MySQL database, including:

- Average skill count grouped by experience range (GROUP BY, AVG, COUNT)
- Most commonly demanded skills overall (GROUP BY skill_name, ORDER BY frequency)
- A JOIN across postings and skills to examine skill demand specifically within entry-level postings
- A CASE-based bucketing of postings into Entry / Mid / Senior seniority bands, compared using GROUP BY and HAVING

2.4 Inflation Scoring (Python)

Using pandas, an inflation score was computed for every posting by comparing its actual skill_count against the baseline (median) skill count for its seniority band, pulled directly from the SQL results via `pd.read_sql_query()`. Postings scoring meaningfully above their band's baseline were flagged as inflated.

2.5 GenAI Explanation Layer

A chatbot layer takes any flagged (or unflagged) posting and generates a plain-language explanation of the result. The system is retrieval-grounded: the posting's exact skill count, seniority band, baseline, and computed inflation score are retrieved from the SQL-backed pipeline first, and only then handed to a large language model (Llama 3.1-8B, accessed via the free Groq API) with an explicit instruction not to contradict the pre-computed classification. This ordering — retrieve real data, then generate explanation — keeps the chatbot's output grounded in fact rather than the model's own guesswork. A rule-based template function serves as an automatic fallback if the API is ever unavailable, ensuring the demo never fails live.

3. Key Findings

28.8%

of postings were flagged as demanding more skills than typical for their experience level — nearly 1 in 3 postings.

3.1 Skill Demand Barely Changes by Seniority

The most significant finding: average skill count stays almost flat across seniority bands, rather than scaling up with experience as one might expect.

Seniority Band	Avg. Skills Required	# Postings
Entry Level (0–2 yrs)	6.50	7,417
Mid Level (3–6 yrs)	6.48	9,225
Senior Level (7+ yrs)	6.49	2,593

This indicates that entry-level candidates are expected to demonstrate almost the same breadth of skills as a candidate with 7+ years of experience. Skill inflation, in other words, is less about experience scaling and more about flat, undifferentiated expectations across the entire market.

3.2 Worst-Offending Companies (Entry-Level Postings)

Filtering to companies with at least 3 entry-level postings (0–2 years), the highest average skill demand was observed at:

- Times Internet — 8.33 average skills per entry-level posting
- Allegis Group, Flutter, E2F, Vanderlande Industries — 8.00 average skills

3.3 Most In-Demand Skills Overall

Across the full dataset, the most frequently requested skills include Data Analysis, Analytical, SQL, Python, and Business Analyst-adjacent competencies — consistent with the Data Science / Analytics domain covered by this dataset.

4. Further Improvement Areas / Next Steps

- Validate the inflation baseline against real hiring outcome data (e.g. which postings actually convert to hires), rather than posting text alone.
- Expand the chatbot to compare a posting against live, continuously updated market data instead of a static historical dataset.
- Package the tool as a browser extension for direct use on Naukri or LinkedIn job pages.
- Standardize skill names (case and synonym matching, e.g. 'Data analysis' vs 'Data Analysis') for cleaner, deduplicated aggregation.
- Incorporate salary data more directly once a larger share of postings disclose compensation (currently ~90% list 'Not disclosed').

5. Justification of the Overall Project

This project addresses a problem that is widely felt but rarely quantified with real data. Its value is twofold:

- For candidates: reduces false self-rejection caused by postings that appear more demanding than the role actually requires, particularly benefiting freshers and early-career applicants.
- For recruiters and companies: highlights where their own job postings may be filtering out qualified candidates for the wrong reasons, offering a concrete, data-backed signal for revising requirements.

Technically, the project demonstrates the full breadth of the competition's requirements — a normalized relational database and real SQL analysis (joins, aggregations), a Python-based scoring pipeline, and a retrieval-grounded GenAI layer that explains results rather than merely presenting numbers.

“One in three job postings ask candidates for more than the market actually expects — JobFair AI gives candidates a data-driven way to tell the difference.”